Persistent Information and the Constrained Future

An IOTA-Grounded Critical Synthesis of the Information–Constraint Principle

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Manuscript draft — 21 July 2026

Abstract

The IOTA research program begins from an information-first substrate hypothesis: present information is the causally retained result of interaction history, embodied in current state and relational organization, and information changes when interaction changes the structure of reachable futures. The Information–Constraint Principle (ICP) proposes that physically effective information is persistent, embodied constraint on an explicitly declared family of possible or probable future responses. This article has two purposes: to audit whether IOTA/ICP is scientifically viable in light of current theory, experiment, close prior art, and project results; and, if so, to specify the research path from constraint persistence to coherence, quasiparticles, fields, and emergent mass and energy. We conduct a critical integrative synthesis rather than a systematic review, separating established field results, project constructive evidence, cross-domain inference, and ontology-level hypothesis. Control and viability theory, computational mechanics, process tensors, counterfactual semantic information, feedback thermodynamics, decoherence, biological memory, and emergent quasiparticle physics establish strong native precedents for parts of the proposal. They also impose a severe novelty test: generic claims that information changes future possibilities are already covered by mature formalisms. IOTA contributes bounded demonstrations of constraint transport through disjoint carrier replacement, prospective lineage succession, finite opportunity lifetimes, and realization redundancy. A new exact specialization identifies its nested-uniform support reduction with forward Kullback–Leibler divergence. This supplies a formal constraint diagnostic and a candidate operator bridge to Gravity from Entropy's log-determinant construction, while leaving the operator map, continuum limit, and any distinctive cosmological prediction unproved. A separate registered construction recovers a bounded one-particle Dirac role, but supplies the quantum substrate, clock, Born readout, and mass parameter. Our verdict is positive but graded. IOTA/ICP has real scientific legs as a coherent, falsifiable research program and operational synthesis, and the project has demonstrated beginnings of its proposed persistence mechanism. It has not yet established a distinct fundamental law, an IOTA-to-quantum derivation, or an information-first ontology. The decisive next bridge is a minimal persistent coherence that yields quantum composition and probability, then a manipulable quasiparticle role, without importing the target structure.

Verdict in one sentence. IOTA/ICP is viable enough to pursue and specific enough to fail, but the present evidence supports a research program and bounded mechanism—not yet a fundamental ontology or law.

1 The persistence problem

The motivating question is elementary to state and difficult to answer without slipping between meanings of information: how can the consequence of an interaction persist after the object or arrangement that first carried it has changed or disappeared? A footprint can outlast a foot; a measurement record can be copied into an environment; angular momentum can be transferred among spin, orbital, and field degrees of freedom; a biological response can survive component turnover. None of these examples requires information to float free of physical realization. They instead distinguish persistence beyond an *original* carrier from existence without *any* carrier.

IOTA takes that distinction as a research problem rather than a slogan. Its organizing hypothesis is that information is the interaction-caused change in the reachable. In condition form, the information present at time t is the part of realized interaction history that remains causally active in present local state and relational topology. In change form, an interaction is informational when it changes which future trajectories or responses are reachable, or changes their probabilities. The proposal is intentionally stronger than Shannon information as uncertainty and narrower than the claim that every physical difference is information.

This paper gives the synthesis two jobs. The first is diagnostic: does this program have genuine scientific legs when compared with the best existing science, including theories that may already contain its central object? The second is constructive: if it does, what sequence of results would move it from a bounded model of persistent constraint

toward an account of familiar physics? The review is therefore allowed to return a negative verdict. A literature collection selected only because it sounds information-friendly would defeat both purposes.

1.1 IOTA as substrate hypothesis, not a frozen cosmic lattice

The IOTA substrate hypothesis treats informational events or distinctions, relations, and local rules of lawful succession as the irreducible explanatory layer. The concrete lattice in the present simulation is a candidate realization, not a claim that its dimensionality, neighborhood, scheduler, or constants are the uniquely correct microscopic architecture of nature. This distinction is essential. A literal fixed lattice inherits immediate burdens: preferred structure, continuum and symmetry recovery, dependence on update order, and the possibility that every reported coherence is a special feature of one engineered rule.

At the theoretical level, write the retained present condition as

$$H_{\leq t} \longrightarrow M_t = (X_t, G_t), \tag{1}$$

where $H_{\leq t}$ is realized interaction history, X_t the current local state, and G_t the active relational organization. Fixed law, topology, scheduler, boundary, constants, and exogenous inputs must be declared rather than hidden in the word "system." Two histories are operationally equivalent at t only when no allowed future intervention distinguishes the retained states at the chosen scale and precision.

The ontology claim is a postulate under investigation: familiar physical objects and quantities are proposed to be stable effective organizations of this underlying structure. It is not inferred from the mere use of a computer model, nor from the truism that physical theories can be encoded as data.

1.2 ICP and the emergence ladder

ICP is the proposed principle connecting the substrate hypothesis to dynamics:

> **Physically effective information is interaction-formed, embodied constraint on an explicitly indexed family of reachable or probable futures; it persists when that constraint role is lawfully re-instantiated beyond its original carrier.**

The proposal has a definite explanatory order:

1. local distinctions and relations make some successor events lawful and others unavailable;
2. interaction changes that successor structure;
3. some organizations preserve and transmit selected constraints;
4. carrier succession allows a relational or causal role to outlast its original realization;
5. persistent coherences may acquire stable collective invariants;
6. suitable coherences may realize quasiparticle and field roles; and
7. mass, energy, geometry, clocks, and macroscopic physics may appear as effective properties of those organizations.

Only the beginning of this ladder has project evidence. The current quantum Dirac construction starts higher up: it assumes quantum amplitudes, composition, a Born readout, a clock, and a mass schedule, then tests whether specified local updates recover a bounded known role [1]. It is evidence that a local construction can realize that role under supplied quantum structure, not yet a derivation of quantum quasiparticles from the base IOTA lattice. The gap is not a footnote. It is the central research frontier.

1.3 Why the synthesis must be adversarial

The phrase "constrains the future" is dangerously easy to satisfy. Given a complete physical state and law, virtually every state difference can alter a future trajectory. Control theory already studies reachable sets; statistical models already condition future distributions; causal models already compare interventions; and semantic-information work already erases correlations and measures effects on later viability. If ICP merely renames these objects, it may be useful language but is not a new physical principle.

The burden is therefore not to find fields in which constraints matter. It is to determine whether the combined requirements of historical formation, physical embodiment, sufficiency for a declared intervention-response family, causal carrier manipulation, succession across original-carrier replacement, and representation robustness produce a new invariant, compression, prohibition, or prediction.

2 Method and evidence discipline

This article is a *critical integrative synthesis and operational proposal*. It is not a systematic review or a claim of exhaustive retrieval. The project registry records the query strata, primary-source checks, source notes, competing explanations, falsification criteria, and a claim–evidence matrix. Searches were expanded adversarially whenever a field offered a likely redundancy account. The final bounded closure covered classical reachability/viability, causal and semantic information, feedback information thermodynamics, positive and critical Quantum Darwinism, quantum reference frames, biological organization and carrier interventions, and emergent quasiparticles. The bounded cross-domain search cutoff was 20 July 2026. A targeted 21 July amendment added two primary Gravity from Entropy sources and the formal IOTA–GfE bridge discussed below; it did not reopen the other search strata or convert the review into an exhaustive search.

Claims are assigned to five evidence strata:

1. **Established field result**: a theorem, experiment, or formal construction within its native assumptions.
2. **Project constructive evidence**: a frozen or prospectively registered construction within a declared model.
3. **Project empirical/computational evidence**: measurements on an engineered or observed data-generating system, without reclassifying that system as fundamental physics.
4. **Synthesis inference**: a cross-field structural mapping that may be supported even when no source states it.
5. **IOTA/ICP hypothesis**: a proposed universality, emergence, or ontology claim.

Publication and peer review are relevant quality signals, not truth labels. The IOTA and Dirac manuscripts are treated as project evidence; the PSA semantic-field manuscript is under review. None is counted as independent replication. Positive, negative, stopped, censored, and withdrawn project results are retained because a viability audit cannot be built only from successful gates.

2.1 Indexed future-law families

For a candidate retained state or record m, intervention family Π, observable algebra $\mathcal{A}$, boundary and resolution declaration B, resource declaration C, and horizon τ, define

$$\mathcal{E}_\tau(m;\Pi,\mathcal{A},B,C) = \left\{P^\pi_{\mathcal{A}}\big(F_{t:t+\tau} \mid m,B,C\big) : \pi \in \Pi\right\}. \tag{2}$$

The corresponding reachable support for policy π is

$$\mathcal{R}^\pi_\tau(m) = \operatorname{supp} P^\pi_{\mathcal{A}}\big(F_{t:t+\tau} \mid m,B,C\big), \tag{3}$$

with topology and tolerance specified for continuous spaces. A single continuous trajectory may have probability zero without being impossible.

Equation (2) is not itself an ICP law. It is shared comparison language. A *constraint* requires a matched contrast between families, for example a natural history-formed carrier m and a reference m_0 prepared with the same declared ordinary state variables and resources:

- a **hard constraint** changes reachable support on a common path or outcome space;
- a **soft constraint** changes weights while support may remain fixed; and
- an **operational constraint** changes a reachability or convertibility preorder under the same allowed operations.

No universal sign for an entropy difference follows from these definitions.

2.2 A relative-entropy constraint diagnostic

For a declared comparison m versus m_0 at x, one distribution-sensitive member of the future-family comparison is

$$\mathfrak{C}^{\pi}_{x,\tau}(m\|m_0;\mathcal{A},B,C) := D_{\mathrm{KL}}(P^{\pi}_{\mathcal{A}}(F_{t:t+\tau} \mid m,B,C)\|P^{\pi}_{\mathcal{A}}(F_{t:t+\tau} \mid m_0,B,C))\,, \tag{4}$$

provided both laws use the same outcome space, intervention, horizon, measure, and coarse-graining, with the first law absolutely continuous with respect to the second [2]. This is an indexed diagnostic, not the definition of ICP: the full family in Eq. (2) retains outcome identity, topology, interventions, and operational order that a single scalar discards.

The IOTA support quantity has an exact specialization. If the post-intervention law is uniform on $\mathcal{R}^{\mathrm{post}}_x \subseteq \mathcal{R}^{\mathrm{prior}}_x$ and the reference law is uniform on $\mathcal{R}^{\mathrm{prior}}_x$, then

$$\begin{aligned} D_{\mathrm{KL}}(U^{\mathrm{post}}_x\|U^{\mathrm{prior}}_x) &= \sum_{f\in\mathcal{R}^{\mathrm{post}}_x} \frac{1}{|\mathcal{R}^{\mathrm{post}}_x|}\log\frac{|\mathcal{R}^{\mathrm{prior}}_x|}{|\mathcal{R}^{\mathrm{post}}_x|} \\ &= \log\frac{|\mathcal{R}^{\mathrm{prior}}_x|}{|\mathcal{R}^{\mathrm{post}}_x|} = \Delta I_x. \end{aligned} \tag{5}$$

This direction is essential: reversing the divergence is infinite under strict support pruning. Non-nested or support-creating interventions require an explicitly chosen alternative treatment rather than silent reuse of Eq. (5) [3].

2.3 The persistent-carrier criterion

A proposed informational carrier M passes the current ICP criterion only if four gates are met:

Formation. Controlled histories create the candidate distinction.

Sufficiency. The distinction is sufficient for the declared future-law family to a frozen error tolerance.

Carrier intervention. Ablating, replacing, transplanting, or scrambling the proposed carrier changes a preregistered component of the future family, while matched shams and resource controls do not.

Robustness. The result survives adequate state augmentation and reasonable representation, partition, and coarse-graining changes, or its representation dependence is stated as a boundary.

Persistence across carrier succession adds one further demand. Let $c_0, c_1, \ldots$ be physical realizations at successive times. They instantiate the same role only relative to a declared equivalence relation $\sim_{\Pi,\mathcal{A},B,C,\tau}$ when replacement preserves the relevant future-response family or a registered relational invariant. Material disjointness alone does not establish sameness of role; sameness of an analyst's label does not establish physical transport.

2.4 Novelty and failure tests

The strongest null is the complete-state ordinary-dynamics account: once every causally relevant variable, law, input, boundary, and control is included, there is no additional historical variable. State augmentation, predictive causal states, process tensors, control/viability models, and counterfactual-semantic measures are field-specific versions of this challenge.

ICP earns distinct physical content only by delivering at least one of the following beyond the best native model:

1. a minimal sufficient physically embodied equivalence class;
2. a representation-stable carrier-succession invariant;
3. a new impossibility or conversion theorem;
4. a demonstrably superior compression of intervention-response families; or
5. a novel, preregistered physical prediction.

Repeated disappearance under adequate state augmentation, failure of matched carrier interventions, dependence on one convenient representation, or the need to import the target quantum structure would lower or defeat the relevant claim.

3 What the project has demonstrated

Project evidence matters because IOTA is not only an interpretation of the literature; it is an attempt to construct and intervene on the proposed mechanism. Its evidential role must nevertheless remain bounded. A simulation can establish consequences of its frozen rules and the realizability of a formal mechanism. It cannot, without an independently validated bridge, establish that nature uses those rules.

3.1 Constraint transport and succession in IOTA

The foundational IOTA result concerns a relational role rather than transport of a marked payload [3]. Thirty-six previously identified bridge-relocation events were replayed; twelve had a role-bearing predecessor bridge, and three events across two seeds passed the strict test. In each positive, tracked regions first lost connectivity and then regained it through a causally necessary successor bridge with no cells in common with the predecessor bridge. Removing that successor destroyed the restored relation. Payload remained zero and the declared conservation ledger was exact.

The result supports a narrow statement: within this event class, a connectivity constraint can be re-instantiated through complete replacement of its local bridge. It does not show indefinite persistence, general carrier independence, or correspondence with a particle or field. Follow-up tracking found finite address-based relation lifetimes of 9, 3, and 51 recorded offsets. A prospective expression-lineage rule extended two relations past original-address death to 170 offsets and at least 601 offsets; the third did not propagate.

Event-resolved interventions then separated route availability from realization. Permitting a selected failed gate rescued six of six same-source gate cases and eight of eight cases in a prospectively screened independent geometry. Preventing source exhaustion reopened all 30 selected same-source routes, but only 28 eventually handed off; an independent exhaustion case also reopened the route without final rescue. An available future was therefore causally relevant but not sufficient. Timing and competing events could close the opportunity before realization.

Necessity tests revealed hidden redundancy. Among 164 factual branches with one observed child, vetoing that child produced an alternate in 122 branches and no alternate in 42. Sequentially suppressing newly realized successors yielded an observed policy cut in 143 branches by depth eight; 19 remained right-censored above that depth and two retained the original source at the horizon. An independent cohort of 79 branches across four new source geometries produced 68 observed cuts, ten right-censored continuations, and one source-survival censoring. These are adaptive, history-changing policy cuts, not a static count of pre-existing interchangeable children.

The evidence record also contains informative limits. Exact precursor placement was causally necessary for later articulation in 25 selected channels, with state-memory dependence in seven, but vetoing the first rebridge never sustained the preregistered 128-offset separation. A proposed rooted graph-role signature showed a directional contrast yet failed its full confirmation criteria ($p = 0.0551$ for the primary paired test and a decisive failure of the separate natural-closeness gate). A subsequent marked topology-by-state instrument was feasible in development, but the independent cohort retained only 16 channels against 20 required; no outcome was opened. The proper statuses are negative for the frozen graph-role representation and untested for the independent repair-affordance hypothesis.

Together, these results demonstrate beginnings of the proposed mechanism: relational constraint replacement, lineage succession, finite opportunity, causal realization, and temporal redundancy occur under frozen local rules and prospective interventions. They do not demonstrate a substrate-independent law.

The revised IOTA formalism also makes one exact mathematical connection. For uniform laws on nested viable sets, its support log-ratio ΔI_x is precisely the forward relative entropy in Eq. (5). This upgrades a verbal entropy resemblance to a proved identity inside the declared finite model. It does not show that IOTA's viable-set measure is geometric, that nature realizes the model, or that the scalar captures cardinality-preserving changes in future-law weights.

3.2 Bounded Dirac role recovery

The quantum-Dirac project asked whether a declared one-dimensional Dirac operational role could be recovered across two separately registered local quantum updates [1]. An ordered product walk and a symmetry-derived direct quantum cellular automaton were tested with frozen input sectors, mass and potential interventions,

refinement ladders, observable witnesses, causal cones, symmetry checks, resource counts, single-use confirmation, and algorithmically separate numerical reconstruction.

The intact evidence supports qualified constructive sufficiency for the shared bounded role core. A hostile post-result audit found that both primary and replay implementations used the same factor-of-two error in an empirical zone-edge predicate, selecting an empty sector. Those values and pass flags were withdrawn; the independently derived analytic tail bound used the correct sector. The audit also established that the free one-excitation blocks are exactly related by a half-cell time-frame conjugacy after mass-angle remapping, and that the reported resource difference depends on nonminimal gate-fusion conventions. Thus the result does not establish two algebraically independent architectures or a hardware advantage.

Most importantly for the emergence ladder, the construction supplies a quantum Hilbert space, Born readout, clock, target map, intervention class, and mass schedule. It shows that explicit local quantum mechanisms can realize the bounded role; it does not yet show an IOTA coherence becoming a quantum quasiparticle or mass emerging as an invariant. This is a successful rung and a precise boundary, not a completed derivation.

3.3 PSA as project evidence

The submitted PSA semantic-field study constructs text signatures from relations to a fixed catalog of propositional anchors [4]. In one embedding model, 115-dimensional signatures retain substantial neighborhood structure from a 4096-dimensional space. Disjoint designed anchor catalogs produce correlated distance geometry beyond an error-independence floor; random projections preserve more raw distance structure but not the same shared distortion. A calibrated projection supports auditable semantic blocking, while negative controls show that proximity does not establish meaning equivalence.

PSA is relevant to ICP because it demonstrates a relational, frame-indexed measurement architecture, tests cross-frame stability, and exposes how a representation can constrain accessible comparisons. It is not empirical fundamental physics. Its objects are sentences, embeddings, anchors, and an engineered retrieval task. The evidence supports a methods analogy and a project example of relational sufficiency and representation control; it does not support the claim that the physical universe is semantic or that the embedding space is an IOTA substrate.

This boundary also corrects an earlier Zipf interpretation. A hot-cell ceiling in an adaptive voxel pipeline was removed by common-level reconstruction; a pipeline-induced count feature cannot be promoted into a mechanism. The PSA lesson retained for ICP is methodological: define the representation and selection process before interpreting a regularity.

4 Closest native formalisms and the redundancy test

The most important literature for ICP is not the literature that sounds most similar. It is the literature capable of making the proposal redundant. Four clusters cover much of the generic future-constraint language: control and viability, empowerment and causal information, constructor and predictive-state theory, and feedback information thermodynamics.

4.1 Reachability and viability already describe constrained futures

Hamilton–Jacobi reachability represents controlled future sets directly. Mitchell, Bayen, and Tomlin characterize a finite-horizon backward-reachable set for a nonlinear differential game as a sublevel set of a viscosity solution to a Hamilton–Jacobi–Isaacs equation [5]. Viability theory identifies states from which at least one admissible evolution can remain inside a declared constraint set, with convergent approximation schemes under stated conditions [6]. These are not loose analogies to Eq. (2); they are mature mathematical theories of accessible and survivable futures under controls and disturbances.

The distinction available to ICP lies upstream of the computed set. A reachability calculation is given the state, law, targets, controls, disturbances, boundary, and horizon. It need not ask how a particular constraint role was formed, where it is physically realized, or whether it can survive replacement of its first witness. If ICP stops at "the state changes what is reachable," it is standard control language. If it can identify a minimal role that is formed and transported across realizations, it may add a new state-equivalence or mechanism question to that language.

4.2 Empowerment and future-option measures

Empowerment makes the overlap still sharper. Its original finite-step form is the channel capacity from present action sequences to later sensed states—a scalar measure of potential perceivable control [7]. Process empowerment replaces fixed open-loop scripts with feedback-conditioned probing policies and can expose controllability hidden by noise or traps [8]. Causal entropic forcing instead builds a finite-horizon path-entropy objective into an effective force and produces adaptive-looking behavior in simple simulations [9].

These works defeat any claim that ICP discovered the idea of valuing or counting future options. They also clarify what ICP does *not* assert. The principle does not presently say that systems maximize reachable-set size, empowerment, entropy, or optionality. A constraint may reduce, redirect, or reorganize futures; persistent structure can increase viable options at one scale while excluding paths at another. Scalar option values also collapse outcome identity: equal capacities need not imply equal future-law families.

4.3 Causal information and the closest semantic rival

Causal-information measures require post-intervention distributions or a defended causal model. Ay and Polani formalize information flow using causal rather than merely observational independence [10]. Counterexamples show why transfer entropy and related dyadic quantities can both overestimate flow and underestimate influence [11]. Predictive association, including a Zipf law or high mutual information, is therefore not a carrier test.

Kolchinsky and Wolpert provide the closest mature formal rival to the phrase "information that makes a difference." They define semantic information as system–environment correlation whose controlled scrambling lowers a declared future viability [12]. The intervention, system boundary, ensemble, horizon, syntactic measure, and viability function are constitutive. On a survival or self-maintenance task, an ICP analysis should specialize or compare with this framework, not silently rename it.

ICP is broader only in a limited sense: its target may be any registered future-response role, not necessarily viability, and it emphasizes formation and succession across original-witness replacement. Breadth is not novelty by itself. Unless carrier succession produces a new invariant or empirical prediction, semantic information and ordinary causal-effect analysis may already exhaust the phenomenon.

4.4 Constructor theory and predictive-state theory

Constructor theory supplies a law-level language of possible and impossible repeatable tasks and characterizes information variables through physical task repertoires such as copying, permutation, and distinguishability [13, 14]. This is the cleanest neighbor for hard physical possibility. It does not, in its basic task definition, supply a history-conditioned stochastic process or require that a present constructor was formed through carrier succession. ICP may eventually add a temporal formation and inheritance layer; for now the relationship is structural, not derivational.

Computational mechanics begins from histories rather than tasks. Causal states are equivalence classes of pasts with the same conditional distribution over futures and yield a minimal optimal predictive representation under the stated process assumptions [15]. Input–output epsilon-transducers extend the construction to conditional output laws under future input sequences [16]. This is almost exactly the $H \to M \to P(F \mid M)$ part of ICP.

The difference is physical, not algebraic. An inferred causal state need not be localized, manipulable, or Pearl-causal; conditioning on an input sequence is not automatically a surgical intervention. ICP's carrier criterion asks whether a physically embodied realization is necessary and sufficient for the relevant family and whether its replacement preserves the role. If that physical test adds nothing to the predictive equivalence class, computational mechanics is the more mature account.

4.5 Feedback information thermodynamics

Feedback thermodynamics gives direct evidence that a retained measurement can change a later physical transformation. Sagawa and Ueda's generalized Jarzynski relation includes information and feedback efficacy in work bounds [17]; Toyabe and colleagues tested information-to-free- energy conversion with position-conditioned feedback on a Brownian particle [18]. An autonomous bit-tape demon makes the memory carrier and its thermodynamic accounting explicit [19]. Current work also shows that finite memory and a limited protocol repertoire alter the optimal feedback value [20].

A 2026 experimental preprint provides an especially relevant boundary case. Time-delayed measurements of a colloid in a viscoelastic bath reveal memory not captured by the visible instantaneous coordinate, and feedback can extract work from hidden environmental degrees of freedom [21]. This is strong evidence that information can persist outside the original or currently observed bearer while remaining physically realized elsewhere. It also supports the state-augmentation null: the enlarged system is Markovian and the hidden modes carry ordinary excess free energy. As a preprint, it remains frontier evidence.

Information thermodynamics therefore supports causal utility and closed physical accounting. It does not support the inference that energy emerges from information, because energy, Hamiltonians, baths, memories, and protocols enter the theory as supplied structure. IOTA must derive mass- and energy-like invariants at a later rung rather than cite feedback conversion as that derivation.

4.6 Result of the redundancy audit

The generic ICP proposition is largely preempted. Mature theories already establish that present physical states, controls, memories, and correlations differentiate future possibilities and performance. The viable novelty target is narrower:

> An interaction forms a physically embodied constraint role; the original witness ceases to instantiate it; a successor realization preserves the registered intervention-response role; and perturbing that successor changes the future family under matched controls.

That target aligns with the strongest IOTA evidence and remains open as a general principle. It is also falsifiable by complete-state reduction, failure of role equivalence, or representation dependence.

5 Quantum records and the emergence bridge

Quantum theory supplies the sharpest positive precedents for information leaving its first bearer, and the sharpest warning against treating that precedent as a derivation of quantum theory. Once a Hilbert space, a tensor or algebraic decomposition, a state, and an intervention class are supplied, standard quantum mechanics already explains how interactions distribute correlations, how some observables become stable under environmental monitoring, and how collective excitations acquire particle-like roles. The IOTA hypothesis begins one level earlier. It must explain how its local informational distinctions could acquire the composition, phase, probability, and transformation structure presupposed by those results.

Table 1 makes this separation explicit. The first rung, and precursor aspects of the second, have bounded IOTA evidence; no project result yet establishes quantum coherence. The current Dirac construction occupies a higher, supplied-quantum rung. The remaining arrows are research hypotheses, not consequences of similarity in vocabulary.

5.1 Multi-time quantum memory is a native future-law formalism

Established field result. The process-tensor framework represents the statistics of a quantum process under a sequence of possible interventions. It separates the experimenter's operations from the multi-time object that maps those operations to outcome probabilities; quantum Markov order and memory classifications then depend on the chosen instruments and system–environment split [22–25]. Recent process tomography demonstrates that such objects can be reconstructed in controlled devices, subject to finite-dimensional, sampling, and control assumptions [26]. This is close to Eq. (2): the future object is not one unconditioned state but a family of responses to allowed operations.

Synthesis inference. A multi-time process is a natural quantum benchmark for an ICP carrier claim. Two histories can leave the same measured reduced system state yet differ in environmental correlations and therefore in their later intervention-response families. The record has not escaped physics; it resides in an enlarged joint state or process. Consequently, state enlargement is both a positive method for locating memory and the strongest null against a new historical variable.

ICP hypothesis. The stronger claim is that a compact, physically identifiable role can be formed, edited, and transferred while preserving the relevant process family. That claim would require histories matched on the best available present state, randomized preservation or rewriting of the proposed carrier, resource-matched shams,

Table 1: The emergence ladder and the evidential status of each bridge.

Rung	Present support	Missing discriminating result
Local interaction → transported constraint	Project construction: selected connectivity roles are re-instantiated in disjoint successors, with causal intervention gates.	Replication across rules, schedules, topologies, and physically equivalent representations.
Transport → persistent coherence	Project lineages sometimes outlive original addresses, a pre-quantum persistence result; quantum and biological records are independent precedents.	A minimal compositional invariant whose persistence is not an artifact of a chosen tracking rule.
Coherence → quantum structure	No base-IOTA derivation. Process tensors, decoherence, reference frames, and the Dirac project all assume quantum theory.	Derive state composition, reversible/interventional dynamics, interference, and probability calibration without importing amplitudes or the Born rule.
Quantum structure → quasiparticle	Established local quantum models and experiments realize fractionalized and topological excitations.	Create and manipulate an IOTA-derived excitation with declared dispersion, statistics, symmetry, conservation, and scattering.
Quasiparticle → fields, mass, and energy	One-dimensional quantum cellular automata recover bounded free-field roles; the project recovers a bounded one-particle Dirac role.	Interacting multiparticle limits, no-go closure, and derived rather than scheduled mass- and energy-like invariants.

and held-out multi-time interventions. If the augmented standard process tensor explains every effect without a new invariant or compression, ICP remains a useful organization of ordinary quantum memory rather than an additional quantum law.

5.2 Environmental records: a real mechanism with an objectivity ladder

Established field result. Decoherence disperses phase information into correlations with an environment, and environment-induced superselection identifies observables comparatively stable under the specified interaction. Quantum Darwinism adds a distinct condition: many disjoint fragments may carry information about a common preferred observable. Environment-as-witness constructions and a broad finite-dimensional channel theorem show how fragment channels can become approximately measure-and-prepare channels for one common POVM [27, 28]. Controlled photonic and spin experiments have observed redundant imprints under engineered or well-characterized conditions [29–31].

This is direct evidence that an interaction-generated distinction can be embodied in secondary degrees of freedom and read away from the source. It is not a demonstration of information without a carrier. Nor do the familiar levels collapse into one another:

$$\text{decoherence} \not\equiv \text{pointer selection} \not\equiv \text{redundant encoding} \not\equiv \text{accessible consensus} \not\equiv \text{objective broadcast record}. \quad (6)$$

Spectrum broadcast structure supplies a stringent state-level form in which multiple fragments have distinguishable conditional supports. Strong quantum Darwinism additionally requires locally accessible classical information, and the multipartite-independence condition must be stronger than pairwise conditional independence [32–34]. Mutual-information plateaus, redundancy, and observer consensus can therefore disagree [35]. Classification results further show that record proliferation requires dynamical structure; arbitrary coupling does not automatically make an objective record [36].

Synthesis inference. Quantum Darwinism supplies the clearest independent model of persistence beyond an original bearer: correlations generated at a source can remain available in other physical carriers. It does not yet supply the ICP succession criterion. Most experiments ask what can be inferred from a fragment now, not whether erasing, swapping, or replacing that fragment later changes a registered future process while all ordinary resources are controlled.

Factorization boundary. The words "system," "environment," and "fragment" presuppose a decomposition. The same global state can display redundant imprints of noncommuting observables under different, non-comparable environment partitions [37]; a fixed Hamiltonian can also admit multiple tensor-product structures with robust

pointer states that need not track familiar locality [38]. This does not undermine an operationally fixed laboratory partition. It blocks the stronger inference from a redundant record to a unique, observer-independent carrier or classical world. An ICP quantum claim must therefore state the factorization or observable algebra, accessible fragment operations, reference resources, and the transformations under which carrier identity is expected to survive.

5.3 Relational facts and quantum reference frames are not memories

Relational quantum mechanics (RQM) and quantum reference-frame (QRF) theory share an insistence that descriptions are relational, but they answer different questions. RQM is an interpretation in which variable values are facts relative to interactions; later work distinguishes approximately stable facts from generic relative facts [39, 40]. Whether its commitments permit consistent cross-perspective fact assignments remains actively disputed [41, 42]. That dispute is interpretive and supplies no independent experimental confirmation of IOTA.

QRF formalisms instead transform states, observables, dynamics, and measurements relative to physical quantum systems. Superposition and entanglement can be frame-relative while consistently transformed operational probabilities remain invariant [43]. Operational equivalence and reversibility depend on the framework, localization, symmetry constraints, and access to global charges [44, 45].

Synthesis inference. This literature requires ICP to distinguish a changed description from a changed future. A frame transformation may relabel subsystems and entanglement without changing any invariant probability. A relative fact may be created in an interaction without becoming a durable, manipulable memory. Neither implication

$$\text{relational fact} \Rightarrow \text{stored carrier}, \qquad \text{frame-relative state} \Rightarrow \text{different reachable family} \tag{7}$$

holds without an additional physical construction. A proposed carrier must either survive the relevant frame equivalences or be explicitly indexed to a declared frame and observer algebra.

5.4 The Dirac construction is a rung, not the missing arrow

The bounded Dirac result belongs between supplied quantum structure and an effective one-particle role. Its local update specifications, convergence tests, causal cones, interventions, and observable witnesses show how a known role can be recovered constructively [1]. The hostile audit and exact time-frame relation limit the architecture-independence claim, but do not erase the intact shared role core.

Its limitation is decisive for the present paper: complex amplitudes, Hilbert composition, unitary updates, a Born readout, a physical clock, and a mass schedule enter as supplied structure. Thus the construction answers "can these local quantum updates realize this bounded Dirac role?" It does not answer "can IOTA interactions create quantum state structure?" A visual analogy between an IOTA coherence and a wave packet would not close this gap. The required bridge must recover interference and composition under novel tests, not encode the target equation in a relabeled update rule.

5.5 Quasiparticles establish possibility, not provenance

Established field result. Local quantum organization can produce excitations whose operational identity is not that of any microscopic constituent. Exactly soluble string-net phases support emergent gauge and fermionic roles, and the Kitaev honeycomb model supports fractionalized Majorana and flux excitations [46, 47]. Material measurements provide model-dependent evidence for such fractional excitations, while controlled processors and quantum gases have created and manipulated non-Abelian or anyonized collective behavior [48–50]. These are not merely metaphorical particles: within their domains they have measurable statistics, braiding, correlations, or response signatures.

One-dimensional quantum cellular automata provide an especially close existence precedent. A multiparticle QCA can match its quantum walk in the one-particle sector and approach a free-fermion QFT at long wavelength. The same analysis proves an obstruction for the proposed class in higher dimensions [51]. The positive construction and the no-go result are equally relevant: local updates can recover a field role, but not generically and not without dimensional and structural conditions.

Synthesis inference. These results defeat the blanket objection that microscopic carrier variables and effective particle identity must coincide. They make the later portion of the IOTA ladder logically plausible. They do not identify the IOTA lattice as nature's substrate because every cited example starts with a quantum Hilbert space, Hamiltonian or circuit, selected phase, and ordinary laboratory resources.

ICP hypothesis and gate. The next project object must be more than a long-lived pattern. A prospectively declared excitation should have creation and annihilation operations, finite propagation, dispersion, composition and exchange rules, symmetry and conserved quantities, scattering or perturbative response, and robustness under carrier replacement and representation change. It must then survive the imported-quantum, lattice-artifact, encoding, and single-target nulls. Passing those gates would establish a quasiparticle-like role in the model. Only a prior derivation of the quantum structure would make it evidence for the full substrate claim.

5.6 Causal order is a boundary on the word "future"

The ordinary form of Eq. (2) presumes a declared temporal or operational order. Process-matrix theory and the quantum switch show that quantum operations can be organized without one fixed global order, and causal witness experiments have certified causal nonseparability under trusted device models [52]. Circuit order, process nonseparability, causal-inequality violation, and indefinite order among localized spacetime events are nevertheless distinct claims [53, 54]. A 2026 photonic experiment reported a Bell-like causal-order-inequality violation at 5.7 standard deviations under stated free-intervention and relativistic-separation assumptions, while retaining locality, fair-sampling, free-choice, and setup loopholes [55].

Boundary verdict. A quantum control can condition the family of later operation sequences. That control is not thereby a history-generated record; causal nonseparability is not thereby indefinite spacetime geometry; and neither result derives causality from information. Where order is not fixed, ICP must replace an ordinary time-indexed future with an explicitly ordered process functional. The causal-order literature is therefore a formal boundary condition on the proposal, not evidence for the substrate ontology.

5.7 Quantum-sector conclusion

The quantum literature supports three ingredients: interactions can create physically separated records; record accessibility is relative to an algebra, factorization, reference frame, and operation class; and organized quantum substrates can realize genuine particle-like excitations. The synthesis inference is that persisted information should be treated as an embodied source–carrier–operation relation rather than a free-floating substance. The open IOTA/ICP hypothesis is that such roles can be constructed from the base informational dynamics and can acquire quantum composition without the quantum target being supplied. No result reviewed here closes that arrow.

6 Cross-domain stress tests: statistics, entropy, life, and gravity

The following domains are not four independent confirmations of ICP. They are four ways to try to break it. Zipf statistics test whether a compelling regularity identifies a mechanism. Entropy tests whether one scalar retains the structure of constrained possibility. Biology tests whether history can build and maintain a causally effective record under ordinary material turnover. Gravity tests whether local geometric entropy can be mapped to an operational future constraint without choosing the operands for agreement; black-hole information further tests whether "preserved," "outside," and "recoverable" remain equivalent when observables, factorization, and computational resources become nontrivial. In each case, the native theory carries the scientific result; ICP is at most a proposed mapping until it supplies an additional invariant, prohibition, compression, or prediction.

6.1 Zipf regularity underdetermines its cause

Established field result. A roughly inverse rank–frequency curve can be generated by incompatible mechanisms. Cumulative-advantage and innovation dynamics produce skew distributions [56]; random character strings provide a memory-free counterexample [57, 58]; communicative least-effort models use a signal–referent optimization [59]; broad latent-variable mixtures can yield Zipf and apparently critical behavior without intrinsic critical tuning [60, 61]; and sample-space-reducing processes generate the law through explicitly history-dependent successor

pruning [62, 63]. The same coarse shape can therefore occur with no memory, hidden common causes, optimization, or genuine path dependence.

Rank–count scaling, the distribution of type counts, and vocabulary growth are also different statistical objects. Their exponent relations are only conditional or asymptotic, with logarithmic corrections near the canonical Zipf case [64]. Holding the final count multiset fixed while permuting event order can produce different discovery paths [65]. A second scaling plot is thus not an independent causal witness if it is algebraically or procedurally coupled to the first.

Strongest rival explanations. Finite samples, aggregation, censoring, adaptive preprocessing, topical mixtures, static grammar or network structure, and transient vocabulary growth can all imitate a proposed constraint mechanism. Visual linearity or a high R^2 is not a distributional test. Likelihood-based fitting, absolute goodness of fit, alternative heavy-tail comparisons, dependence-aware uncertainty, and replication are prerequisites even for the modest claim that a power law is supported on a declared range [66, 67].

Synthesis inference. Zipf is useful to ICP precisely because it demonstrates causal underdetermination. The discriminating object is not an exponent but the law of future reuse, novelty, and succession among histories matched on present marginal counts. Within-document shuffles, block shuffles, frozen prefixes, latent conditioning, and held-out prediction can reveal path dependence while preserving a marginal. Those are analytic interventions; they do not yet manipulate a physical carrier.

ICP hypothesis. A sequence-bearing role becomes ICP evidence only if a candidate embodied record is formed, is sufficient for the declared future law, survives rival conditioning, and changes that law when its carrier is erased, exchanged, or replaced under matched controls. If counts, latent variables, and ordinary dynamics absorb the effect, the Zipf result rejects rather than supports the strong claim. No Zipf marginal in the present project establishes ICP.

6.2 Constraint appears in entropy only after the problem is declared

The phrase "constraint lowers entropy" is true in special nested problems and false as a general identification. If $C_2 \subseteq C_1$ are measured by the same μ, then the Boltzmann-style support quantity obeys

$$k \log \mu(C_2) \leq k \log \mu(C_1). \tag{8}$$

Adding a feasible constraint to the same maximum-entropy optimization likewise cannot raise the optimized objective. A physical intervention, however, can also change dynamics, Hamiltonian, reference measure, observable algebra, coarse-graining, operation class, or horizon. Under those changes an entropy may rise, fall, or remain fixed.

Established field results. Shannon entropy quantifies average coding uncertainty for a declared distribution [68]. Boltzmann multiplicity and Jaynesian maximum entropy use closely related mathematics for different scientific jobs: compatible physical microstates versus inference subject to selected expectation constraints [69, 70]. Lieb and Yngvason begin instead with an adiabatic-accessibility preorder and, under equilibrium axioms, derive an entropy representing that order; outside that regime lower and upper entropies can leave genuine incomparability [71, 72]. Here accessibility is logically prior to its scalar representation.

Temporal organization separates one-symbol entropy, entropy rate, predictive information, and predictive-state memory. A fair independent coin and an alternating binary process with random initial phase both have one bit of single-symbol entropy, while only the latter stores a bit that fixes every later symbol. Entropy-convergence and computational-mechanics analyses make this distinction formal [15, 73]. Likewise, topological entropy tracks growth of distinguishable path support, whereas metric entropy depends on the invariant probability measure [74–76]. Hard possibility and soft weighting can therefore vary independently.

Path-level and thermodynamic quantities add other projections. Large- deviation rate functions retain the exponential cost of rare future sectors, but conditioning on a rare future is not a physical intervention that causes it [77, 78]. Fluctuation relations compare forward and reverse path probabilities and work statistics; they do not measure stored predictive organization [79–81]. In a specified driven stochastic model, nonpredictive information is tied to dissipation, a powerful but conditional connection rather than a universal identity [82].

Table 2: Constraint structure and entropy are not one-to-one.

Declared constraint object	Native probe	Information lost by one scalar
Compatible states or paths	Boltzmann/support or topological entropy	identity of excluded states, graph structure, and intervention response
Probability weights	Shannon, Rényi, or metric entropy	outcome identity, causal provenance, and rare-event geometry
Predictive process	entropy rate, excess entropy, statistical complexity	full causal-state topology and physical carrier location
Forward/reverse experiment	entropy production and fluctuation relations	stored memory and support accessibility
Allowed transformations	majorization and resource-monotone families	incomparability and operations hidden by the chosen monotone
Observable algebra	von Neumann or algebraic entropy	distinctions inaccessible outside the selected algebra

Quantum and finite-resource settings make scalar insufficiency sharper. Conditional min- and max-entropies answer operational one-shot tasks at a chosen error tolerance [83]. Pure-state LOCC convertibility is governed by majorization rather than entanglement entropy alone [84]. Thermal resource theories can require a family of generalized free energies, with additional coherence constraints when superpositions across energies matter [85, 86]. Restricting control Hamiltonians changes extractable work while the state remains fixed [87]. In gauge theories even the entropy of a region depends on the chosen observable algebra, center, or edge-mode prescription [88, 89].

Synthesis inference. Constraint is upstream only in a carefully indexed sense: the compatible set, path law, operation preorder, observable algebra, reference state, task, and resource budget are specified first; an entropy is then a functional or monotone of that structure. Equal entropy does not imply equal accessible futures, and lower entropy does not imply more organization. Feedback thermodynamics establishes that a recorded distinction can improve work extraction within ordinary energetic accounting [17, 18]; it does not derive energy from information.

ICP hypothesis. An embodied record may change support, weights, predictive memory, rare-event cost, dissipation, or a conversion preorder. The experiment must say which, then compare the complete native model after carrier intervention and state augmentation. Searching across entropy definitions until one moves in the desired direction would be representation selection, not evidence for a principle.

6.3 Gravity from Entropy supplies a geometric bridge target

Established field result within the GfE construction. Gravity from Entropy (GfE) defines higher-order metric operators, assumes their flattened ratio is positive, and writes its gravitational log-determinant term as

$$\mathcal{L}_{\mathrm{GfE}} = -\operatorname{Tr}_F \ln \mathbf{\Theta}, \qquad \mathbf{\Theta} = \widetilde{\mathbf{G}}\,\widetilde{g}^{-1}, \qquad \boldsymbol{\mathcal{G}} = \mathbf{\Theta}^{-1} \tag{9}$$

[90, 91]. These metric operators are not normalized probability states, and the subscript F denotes the flattened-tensor trace rather than a probabilistic state space. Thus Eq. (9) is not automatically a Kullback–Leibler or Umegaki divergence. In the thermodynamic development, an eigenmode contributes

$$c_k := \frac{\delta s_k}{\delta v} = \ln \mathcal{G}_k, \qquad \theta_k = e^{c_k} - 1, \qquad \frac{\delta \epsilon_k}{\delta v} = e^{c_k} - 1 - c_k. \tag{10}$$

Individual c_k can be signed; a nonnegative mode-by-mode identification with Eq. (4) is therefore generally invalid.

Synthesis inference: the bridge formula. Let Q_x^{prior} and Q_x^{post} be as-yet-undefined positive operators whose flattened determinants encode one declared measure of IOTA reachable futures. The proposed bridge is

$$\begin{aligned} \mathbf{\Theta}_x &\stackrel{?}{=} (Q_x^{\mathrm{prior}})^{-1/2} Q_x^{\mathrm{post}} (Q_x^{\mathrm{prior}})^{-1/2} \sim Q_x^{\mathrm{post}} (Q_x^{\mathrm{prior}})^{-1}, \\ \det_F Q_x &= a\,\mu(\mathcal{R}_x), \qquad a > 0 \text{ common to prior and post.} \end{aligned} \tag{11}$$

Here $\sim$ denotes similarity; the first representative is positive and has the determinant ratio of the original product. For the finite IOTA identity, μ is counting measure. Under these algebraic conditions,

$$\begin{aligned} -\ln \det_F \mathbf{\Theta}_x &= \ln \frac{\mu(\mathcal{R}_x^{\text{prior}})}{\mu(\mathcal{R}_x^{\text{post}})} \\ &= \ln \frac{|\mathcal{R}_x^{\text{prior}}|}{|\mathcal{R}_x^{\text{post}}|} = \Delta I_x = D_{\text{KL}}(U_x^{\text{post}} \| U_x^{\text{prior}}). \end{aligned} \tag{12}$$

For a non-counting base measure, the same determinant algebra would instead define $\Delta I_x^{(\mu)}$ for normalized restrictions of that measure; it cannot silently replace the cardinality-defined ΔI_x. The question mark in Eq. (11) carries the missing physics: the operators, common domain, metric meaning, covariance, functional variation, and continuum regularization remain to be derived from IOTA and GfE.

Conditional scaling and discriminator. If a mode-resolved IOTA contribution $\chi_{x,k}$ and a physical mode dictionary were independently derived, the conditional sectorwise crosswalk would be

$$\begin{aligned} &\chi_{x,k} \widehat{=} c_k := \delta s_k / \delta v, \\ &\mathcal{G}_k = e^{\chi_{x,k}}, \qquad \theta_k = e^{\chi_{x,k}} - 1, \qquad \delta\epsilon_k/\delta v = e^{\chi_{x,k}} - 1 - \chi_{x,k}. \end{aligned} \tag{13}$$

Here $k \in \{0, \ldots, 4\}$ labels the five GfE sectors and z_k is the corresponding degeneracy, with $z_0 = z_1 = 1$ and $z_2 = z_3 = z_4 = 3$. In the reported weak-curvature Friedmann expansion, $u_k = \omega_k H^2$ with $\omega_k = \omega_k' \ell_{\text{P}}^2$, and $\bar{\omega}_{[1]} := \sum_{k=0}^{4} z_k \omega_k$. This cannot be imposed generally because c_k can be signed while Eq. (4) is nonnegative. An aggregate $\chi_x \widehat{=} \sum_k z_k c_k$ also determines neither a single temperature nor $\delta\epsilon/\delta v = \sum_k z_k (e^{c_k} - 1 - c_k)$ without the full mode vector.

For weak-curvature, late-time Friedmann backgrounds, modes with $\omega_k \neq 0$ obey $c_k \simeq \omega_k H^2$ and $\delta\epsilon_k/\delta v \simeq \frac{1}{2}\omega_k^2 H^4$. For non-de Sitter solutions with $H \sim t^{-1}$ these scale as t^{-2} and t^{-4}. The summed entropy's leading coefficient $\bar{\omega}_{[1]}$ vanishes in the reported Milne and $w = 1$ cases, so an aggregate H^2 scaling requires $\bar{\omega}_{[1]} \neq 0$. All of these asymptotics insert standard Friedmann behavior into the GfE functionals; they are not yet a novel ICP prediction or a discriminator from GR/ΛCDM.

A first bridge-specific test is instead to construct Q_x from frozen IOTA microdynamics and $\mathbf{\Theta}_x$ independently from the geometric sector, then test whether their scale-dependent log determinants agree without fitting the equality. Generic branching, percolation, ordinary complete-state dynamics, GR/ΛCDM, and unbridged GfE do not themselves supply that cross-level identity. A cosmological claim requires more: the derived continuum map must produce a transition, correction, or observable that is absent from the named complete rivals and is registered before data comparison. At present this is a precise research bridge, not empirical support for ICP or GfE.

6.4 Biology supplies persistent records, not a substrate ontology

Established field result. Biological constraint-closure theory models relatively conserved structures that act on material processes while depending on one another for maintenance in an open system [92]. The description is scale- and timescale-relative: an enzyme may constrain one reaction while being produced and degraded on another horizon. Whether closure alone grounds function, normativity, or life remains contested; its most defensible role here is a model-guided empirical abstraction rather than a definition of purpose [93–95].

Two experiments closely instantiate the persistence question. In bacterial acquired immunity, phage exposure installs a sequence record in a CRISPR locus, and adding or deleting matching spacers changes later resistance [96]. In yeast, prior inositol starvation creates INO1 transcriptional memory involving a DNA element, nuclear-pore association, histone state, and writer–reader machinery. The H3K4me2 mark persists for roughly four divisions after removal of the initiating factor Sfl1, but is lost when maintenance machinery is removed [97, 98]. The record outlives its initiator; it never becomes independent of all material realization.

Evolutionary experiments show a related path effect. Exhaustive construction of all 32 combinations of five TEM beta-lactamase substitutions found only 18 of 120 monotonic direct paths accessible under the study's measured landscape and strong-selection/weak-mutation assumptions [99]. Ancestral reconstruction of the glucocorticoid receptor showed that rare permissive substitutions made later functional changes tolerable [100]. These are strong demonstrations of context-dependent probability or operational accessibility, not context-free proofs of physical impossibility; population size, drift, recombination, environment, and horizon alter the path graph.

Constructive origin-of-life work supplies both positive and negative controls. Designed RNA enzymes can cross-catalyze exponential replication, and cooperative RNA networks can outgrow selfish cycles [101, 102]. Yet tested compositional autocatalytic networks show that self-maintenance and recurrence need not yield selected heredity or evolvability [103]. Lipid or droplet compartments causally alter co-inheritance, resource sharing, and parasite competition; synthetic protocells can preserve full-length replicators relative to bulk conditions [104]. Recent synthetic cells integrate genome replication and lipid synthesis in partial architectures, while still relying on supplied expression systems, substrates, energy, and externally imposed cycling [105].

Strongest rival explanation. Every example has a sufficient ordinary mechanism: nucleic-acid sequence and base pairing, chromatin chemistry, protein structure and population dynamics, reaction kinetics, diffusion, sequestration, and external laboratory operations. A state description that omits the spacer, mark, enzyme, boundary, or bath will appear history-dependent; an adequate physical state includes the carrier. Moreover, counterfactual semantic information already scrambles system–environment correlations and measures effects on later viability [12], while Causal Leverage Density proposes the corresponding comparison over distributions of future phase-space trajectories [106]. Generic future-shaping is not ICP novelty.

Synthesis inference and ICP hypothesis. Biology strongly supports the architecture

$$\text{history} \rightarrow \text{embodied record} \rightarrow \text{changed later response}, \tag{14}$$

including cases in which the initiating event or molecule is gone. The defensible ICP extension is to add formation, sufficiency, carrier intervention, succession, and representation-robustness tests across a family of challenges. Biology does not establish that information is ontologically prior to the molecules, nor that IOTA produces living organization. It does provide well-controlled physical systems in which the full carrier criterion can be tested prospectively.

6.5 Black-hole information is a frontier test of access

Black-hole theory is relevant because it separates claims often compressed into the sentence "the information comes out." The necessary hierarchy is

$$\begin{aligned}\text{global preservation} \not\Rightarrow \text{algebra-relative distinguishability} \not\Rightarrow \text{logical reconstruction} \\ \not\Rightarrow \text{message recovery} \not\Rightarrow \text{efficient physical recovery}.\end{aligned} \tag{15}$$

Each arrow requires a subsystem or algebra, operation class, code subspace, error tolerance, and resource budget.

Established or controlled field results. Hawking's semiclassical calculation motivates pure-to-mixed evolution after an inaccessible region is traced out [107]. Page's curve is a conditional random-state benchmark: if formation and evaporation are unitary and the bipartite state is typical, the entropy of a chosen radiation subsystem rises and falls [108]. Hayden–Preskill recovery shows rapid message availability under idealized unitary mixing and access to early radiation, while the Harlow–Hayden result separates information-theoretic recoverability from feasible decoding by an exponential complexity barrier [109, 110].

Quantum extremal surfaces and islands extremize generalized entropy, combining an area term with bulk entropy. In controlled AdS/CFT, JT-gravity-plus-bath, braneworld, and related models, saddle transitions reproduce Page behavior and move a logical interior algebra into a radiation entanglement wedge [111–113]. Holographic quantum error correction and bulk–boundary relative-entropy results make redundant reconstruction precise inside a declared code subspace [114, 115]. Replica wormholes supply controlled entropy calculations, with ensemble interpretation and analytic-continuation assumptions kept explicit [116, 117].

These are major theoretical advances. They do not yet provide a unique microscopic carrier, an efficient decoder for a realistic four-dimensional evaporating black hole, or direct astronomical observation of a Page curve or information-bearing Hawking correlations. A replica saddle is a contribution to a calculation, not by itself a real-time memory trajectory or a physical counterfactual branch.

Observable-algebra boundary and current frontier. Gravity resists a naive tensor factorization because gauge constraints, gravitational dressing, and asymptotic charges link nominal regions. One line of work argues that a complete asymptotic algebra can distinguish the state and that the ordinary split property fails [118]; another

constructs proper radiation subalgebras and Page behavior in massless gravity under perturbative assumptions [119]. Recent responses argue that excluding the Hamiltonian or using a blind detector makes the Page curve a statement about restricted access rather than late emergence of information [120]. Complementary 2026 models obtain late saddles or asymptotic recovery in specifically declared massless-gravity or two-dimensional settings, without closing the full realistic evaporation problem [121, 122]. The dispute cannot be settled by calling both sides "constraint"; they choose different observable algebras and therefore answer different operational questions.

Candidate carriers are likewise conditional. High-order radiation correlations, code embeddings, gravitational dressing, soft charges, remnants, and state-dependent interior maps are distinct proposals. Soft charges are real conserved structures, but infrared factorization arguments challenge their capacity to encode hard-state information [123, 124]. Recent island-encoding work gives a concrete Goldstone/Stueckelberg dressing mechanism for massive-gravity bath models, not yet for an isolated astrophysical black hole [125].

Synthesis inference and ICP hypothesis. The black-hole lesson is operational modesty. The same Page-shaped curve can coexist with different channels, algebras, reconstruction maps, carriers, and recovery costs. ICP usefully insists that preservation and accessibility be indexed by these choices. It does not currently solve the information paradox or derive gravity. A genuinely relevant carrier test would vary a bath coupling, radiation erasure, detector algebra, code embedding, dressing, or decoder budget and separately score entropy, distinguishability, reconstruction error, message fidelity, and cost in one fixed microscopic theory.

6.6 Cross-domain result

The four stress tests converge on a negative and a positive conclusion. The negative conclusion is that a Zipf exponent, an entropy value, a persistent biological response, or a Page curve cannot identify a unique informational carrier or mechanism. Native statistics, thermodynamics, molecular dynamics, quantum channels, and gravity remain sufficient explanations within their domains. The positive conclusion is that each field already requires some version of an indexed possibility structure: a path law, operation preorder, observable algebra, carrier-bearing state, or resource-bounded decoder.

That convergence supports ICP as a disciplined comparative research program. It does not establish ICP as a fundamental law. The proposal gains physical content only when an interaction-formed role survives original-carrier replacement and its intervention produces a robust future-law difference not exhausted by the complete native model.

7 Adversarial viability audit

The synthesis can now answer its first question. The answer is neither that information language is empty nor that IOTA has been confirmed as fundamental. The evidence supports a narrower conclusion: IOTA/ICP is a coherent, operationally testable research architecture with bounded constructive results and independent precedents for its components. Whether it is a distinct physical principle remains unresolved.

7.1 The complete-state challenge

The strongest null hypothesis is ordinary and severe. Let Z_t be a complete physical specification at time t, including all degrees of freedom relevant at the chosen precision, their correlations, boundary conditions, accessible resources, controls, and the applicable dynamics. Then the future family is already fixed as

$$\mathcal{E}_\tau = \Phi_\tau(Z_t; \Pi, \mathcal{A}, B, C). \tag{16}$$

On this account, a memory, scar, environmental record, regulatory state, or IOTA bridge role matters only because it is part of Z_t. History is relevant exactly when it has left a present physical difference; enlarging the state until it is sufficient removes any independent history variable. Calling that difference "information" and its effects "constraint" adds a useful description, but no new law.

This null is not defeated by showing that two visibly similar systems behave differently, by finding mutual information between past and future, or by demonstrating that an environmental degree of freedom carries a record. Hidden state, conserved quantities, unmeasured correlations, ordinary free energy, and control mismatch must be exhausted

first. Nor is it defeated by a compact description alone: a predictive statistic may be epistemically economical without being a new physical entity.

ICP becomes distinct only if its carrier-equivalence construction does useful work beyond Eq. (16). Candidate achievements include a minimal physically manipulable sufficient class, an invariant across genuine carrier succession, a new conversion or impossibility theorem, a large and validated compression of intervention–response laws, or a preregistered prediction missed by the best complete-state model. The present project has not yet delivered one at fundamental-physics scope.

The exact IOTA relative-entropy identity and the proposed GfE operator map make the cross-domain claim sharper, but formal coherence alone does not change this verdict. The map must first produce a representation-stable invariant or a preregistered prediction beyond Eq. (16); otherwise it is a calibrated redescription of two native constructions.

7.2 Eight failure modes that remain live

1. **State-augmentation collapse.** A claimed historical effect disappears when the carrier, environment, or multi-time state is adequately represented. This would leave ICP as an organizing vocabulary for ordinary memory rather than an additional state principle.
2. **Representation dependence.** The carrier or constraint exists only under a convenient partition, graph encoding, observable algebra, or coarse-graining. Quantum factorization results make this a substantive risk; the negative IOTA graph-role gate shows it is also a project-level risk.
3. **Inferential–physical equivocation.** An observer's loss of uncertainty is mistaken for a change in the system's physical possibilities. Shannon information, Bayesian conditioning, and a record readout do not by themselves establish causal exclusion or a transported role.
4. **Generic-dynamics redescription.** Any present state constrains its future under a law. Without the formation, intervention, succession, and robustness gates, the proposed principle collapses into this truism or into existing reachability, viability, semantic-information, or predictive-state theory.
5. **Engineered-lattice artifact.** IOTA effects may depend on one update rule, scheduler, topology, boundary, or analyst-defined address scheme. A successful construction establishes possibility within that model, not use by nature or universality across realizations.
6. **Imported emergence.** A calculation can reproduce a target only because quantum amplitudes, a Born rule, a clock, a target Hamiltonian, or a mass schedule were supplied. The current Dirac result openly occupies this category: valuable role recovery, but not base-level emergence.
7. **Summary-statistic underdetermination.** Zipf exponents, entropy values, Page curves, and aggregate correlations can be shared by mechanisms with different carriers and future laws. A suggestive scalar cannot identify the proposed causal architecture.
8. **Self-confirming evidence.** Project papers, simulations, and analysis pipelines may share assumptions and error modes. The disclosed Dirac predicate withdrawal and stopped IOTA confirmation are reasons to preserve adversarial audits and external replication, not reasons to erase the intact bounded results.

7.3 What the evidence does and does not jointly license

Several claims survive when the domains are combined. Interactions can create records in secondary carriers. Present physical organization can alter support, probabilities, viability, convertibility, work extraction, and accessible operation order. Some roles survive turnover of their first local realization. Collective quantum systems can support excitations with particle-like properties that are not properties of their microscopic constituents. These are established field results or bounded project constructions within declared assumptions.

The conjunction does not establish the IOTA ontology. No reviewed experiment shows information existing without physical realization; ICP does not require that stronger claim. More importantly, no current derivation begins with IOTA's base events and relations and obtains quantum composition and probabilities, then interacting fields, mass–energy invariants, geometry, and clocks without placing those structures in the premises. Black-hole

reconstruction, Quantum Darwinism, feedback work extraction, biological organization, and quasiparticles each demonstrate only a segment of that ladder.

The name *Information–Constraint Principle* remains defensible if "principle" denotes the proposed organizing and research principle now under test. It should not be presented as an established law. Renaming it merely to avoid that distinction would not strengthen the science; explicit status labels do.

7.4 Graded verdict

Dimension	Rating	Evidential basis and ceiling
Conceptual coherence	Strong	Retained condition, informational change, intervention, future family, and carrier succession are separable and operationally specifiable.
Operational precision	Moderate–strong	The four carrier gates and matched future-family contrasts can fail; a generally accepted metric and invariance theorem are still absent.
Distinctness from prior art	Partial	Formation plus succession beyond the original witness is narrower than generic reachability, but complete-state, causal-state, process-tensor, and semantic-information accounts remain live.
Project evidence	Moderate, bounded	Frozen IOTA interventions establish constructive examples; Dirac role recovery is intact within its supplied quantum layer. Neither is an independent observation of nature's substrate.
Independent grounding	Indirect, substantial	Physics and biology establish component mechanisms, not their IOTA integration.
Falsifiability	Strong locally	Carrier edits, state augmentation, representation changes, and held-out interventions can reject individual claims; no single finite test yet rejects every possible IOTA realization.
Information-first ontology	Open	No result uniquely selects informational events and relations as nature's substrate or derives all supplied physics.
Research tractability	Strong	The next bridges have concrete observables, controls, nulls, and stopping rules.

Table 3: Viability verdict at the 20 July 2026 bounded-search cutoff, with the targeted 21 July GfE amendment. Ratings refer to the research program, not to the truth of the ontology.

The resulting verdict is positive but graded:

> **IOTA/ICP has real scientific legs as a coherent, falsifiable research program and as an operational synthesis of persistent constraint. Bounded project results demonstrate beginnings of the proposed mechanism. Current science supplies independent precedents for its components, but neither their integration nor an information-first ontology has been established. ICP is therefore a viable proposed principle and research architecture, not an established fundamental law.**

This verdict is more useful than an unconditional endorsement. It identifies exactly what survives current science and the observations that would force a weaker interpretation.

8 Research path: from persistence to physics

The synthesis's second function is to convert the verdict into an ordered program. The order matters. A higher rung cannot be counted as evidence for a lower rung when the lower structure was supplied in advance.

8.1 Rung 1: replicate carrier succession

The immediate goal is not a more elaborate cosmological simulation. It is a harder test of the result already obtained. Repeat the disjoint-replacement and lineage protocols under alternative local laws, asynchronous and randomized schedulers, topologies, boundary conditions, state encodings, and independent implementations. Freeze the role equivalence and opportunity horizon before opening outcomes. Use held-out source geometries and publish null and censored cases. Estimate Eq. (4) under the same frozen choices and require its qualitative result to

survive law, scheduler, topology, encoding, boundary, and coarse-graining changes before treating it as more than a model-indexed diagnostic.

Success would establish that constraint succession is a class of mechanism rather than a quirk of one coordinate system. Failure under modest representation or rule changes would lower the claim to a model-specific phenomenon. Either outcome is informative.

8.2 Rung 2: isolate a minimal persistent coherence

A connectivity role is not yet a quantum coherence. The next target should be the smallest organization whose empirically complete future family cannot be compressed to any one constituent or original address, yet is preserved by a registered succession map. Required measurements include lifetime, propagation, composition, interference-like response, robustness to local perturbation, and a causal edit of the current carrier. Competing ordinary state summaries must be fit on training interventions and evaluated on held-out interventions and horizons.

A decisive result would identify a representation-robust equivalence class of realizations with a common intervention–response role. A decisive negative would show that each apparent coherence reduces to a changing list of local states with no stable role beyond the analyst's tracking rule.

8.3 Rung 3: derive the quantum layer

The quantum bridge must produce, rather than assume, the structures used by the Dirac construction. At minimum it must account for:

1. a state space with a defended composition rule;
2. reversible or appropriately open effective dynamics;
3. interference and contextual probability constraints;
4. a measurement or intervention semantics;
5. the Born probability rule, or a quantitatively distinct replacement;
6. operational clocks and reference-frame transformations; and
7. a controlled continuum or scaling regime.

Merely storing complex numbers on lattice sites would import the target. The derivation must show why the effective equivalence classes and transformation rules arise from base succession and why relevant predictions are insensitive to arbitrary partition or coordinate choices.

The strongest near-term physical analogue is a carrier-edit experiment. Form a record through two matched histories, characterize its secondary physical carrier, then preserve, erase, swap, scramble, or rewrite that carrier while balancing energy, charges, fields, pulses, and environmental correlations. Estimate the later process or reachable family under a common intervention set. If an augmented standard quantum state predicts every result, ICP gains a useful embodied-memory instance but no extra ontology. A preregistered residual beyond that model would materially raise the verdict.

8.4 Rung 4: construct a quasiparticle role

Once a quantum layer exists, the appropriate target is a manipulable collective excitation, not the full Standard Model. The registered role should require: creation and annihilation procedures; localized or relational transport; dispersion; composition; symmetry transformation; a conserved or topological label; controlled scattering; robustness and lifetime; and convergence across scales. Established string-net, topological-order, quantum-simulator, and QCA results prove that such emergence is possible in other local quantum systems. They also supply hostile benchmarks and lattice no-go constraints.

The current Dirac construction then becomes a downstream validation target: a derived excitation should reproduce its bounded operational witnesses without being tuned by an externally supplied Dirac map, mass schedule, or Born readout.

8.5 Rung 5: interacting fields, mass, and energy

Free one-particle kinematics is not a field theory. The next stage must recover many-excitation composition, statistics, interactions, locality or a precise replacement, scattering observables, conservation laws, and controlled renormalization or continuum behavior. Fermion doubling, symmetry recovery, gauge constraints, anomalies, and Lorentz-violation bounds are stopping tests, not optional finishing details.

Only after this stage should the project claim emergence of mass and energy. The target is not an analogy between information and work. It is an effective invariant that plays the mass or energy role across processes: governing dispersion, time translation, inertia, conservation and exchange, and coupling in the appropriate regime. Its numerical value and transformation law must be derived from the organization, not placed in an update angle or external Hamiltonian.

8.6 Rung 6: geometry, clocks, and macroscopic recovery

An information-first theory must ultimately explain why effective causal and metric relations approximate the spacetime structure already tested by physics. The project must identify operational rods and clocks, recover frame-covariant predictions, specify when causal order is definite, and derive the scale at which continuum geometry is accurate. Black-hole information is a late stress test of that theory—especially algebra-relative recovery, generalized entropy, and computational accessibility—not an early proof of the substrate.

Macroscopic recovery also requires thermodynamic and statistical explanation. The theory must say why entropy concepts appropriate to different tasks agree where they should, diverge where they must, and yield ordinary irreversible phenomena without turning one entropy scalar into the definition of constraint. Zipf-like distributions, if they occur, require a mechanism and matched-marginal counterfactual tests rather than a fitted slope.

The GfE bridge in Eq. (11) makes one part of this rung unusually concrete. Construct Q_x from IOTA future laws rather than from the target geometry; prove positivity and covariance; control the finite-size and continuum determinants; and test Eq. (12) across scales and held-out geometries. Only then should the program derive a cosmological correction, transition scale, or observation and compare it with GR/ΛCDM, native GfE, and nongravitational branching or percolation nulls. The immediate goal remains the harder carrier-succession replication in Rung 1, not a premature cosmological simulation.

8.7 Decision gates and stopping rules

The program should continue while each rung yields preregistered constructive results, useful failures, sharper invariants, or discriminating predictions. It should be narrowed or radically reformulated if any of the following persists after technically adequate attempts:

- complete-state augmentation repeatedly removes the candidate historical variable;
- carrier interventions fail under matched physical controls;
- the role disappears under reasonable representation, factorization, scheduler, or coarse-graining changes;
- the quantum bridge requires Hilbert-space, probability, clock, or target-dynamics structure to be inserted by hand;
- higher-dimensional, interacting, symmetry, or continuum tests fail their relevant no-go boundaries; or
- no new compression, invariant, prohibition, or physical prediction emerges beyond native theories.

These are scientific stopping rules, not rhetorical concessions. A result that reduces ICP to a precise theory of physically embodied memory would still be valuable, even if the information-first ontology fails.

9 Conclusion

The original question survives the audit in a sharpened form. Information can persist beyond the object that first carried it, but every established example does so through continuing physical realization in another state, relation, or organization. IOTA has constructed bounded cases in which a relational constraint is destroyed locally and re-instantiated through a causally necessary successor. Established science shows that records can be distributed,

biological constraints recursively maintained, future options formally restricted, and particle-like roles produced collectively. Those results make the larger program scientifically plausible; they do not complete it.

The honest present claim is therefore deliberately finite. IOTA/ICP has begun to demonstrate the mechanism it proposes and has a tractable path on which the next steps can succeed or fail. The next result that would materially change the verdict is not another cross-domain resemblance. The exact nested-support relative-entropy identity and the conditional GfE log-determinant map now turn one resemblance into a formal, falsifiable bridge target, but no physical operator map or novel observation has yet been derived. The decisive next result is still a minimal persistent coherence, derived from the base IOTA layer, that survives carrier succession and supports a held-out quantum intervention role without importing quantum state structure or probability. From there, a quasiparticle is a defensible next target. Until those bridges are crossed, information primacy remains the program's hypothesis rather than its conclusion.

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